# JING-ZHANG GOES FIRST

## GREEN:LIGHT

An urban corridor that gives AI scenarios a green light ・ graded right-of-way ・ a way out

Autonomous delivery, low-speed shuttles, inspection and guide robots, mobile clinics, mobile classrooms, rehabilitation aids — all of these AI scenarios are stuck at the same bottleneck: no street is willing, or ready, to let them out, so they only ever run demonstrations inside closed campuses.

The Jing-Zhang Railway was the first trunk line surveyed, designed and built by Chinese engineers, opened in 1909, carrying coal and people. It left behind a continuous linear green corridor of about 9.72 km running through all three key areas — the only continuous channel in this district that reaches from a trunk line to residential doors.

This proposal turns it into an urban corridor where AI scenarios can take to the street legally, slowly, under supervision and with a way out, assembling in one move the full set of facilities and rules that street entry requires. The green light is a metaphor for admission — and it can always be switched back to red.

### Key metrics

| | |
|---|---|
| Design area | 11.41 sq km |
| Trunk length | 9.72 km |
| Grade A length | 5.30 km |
| Grade B length | 3.31 km |
| Grade C length | 1.11 km |
| Bottleneck share | 11.4% |
| Green ratio | 24.7% |
| To-door points | 32 |
| Housing covered within 300 m | 56.0% |
| Landmarks | 3 |

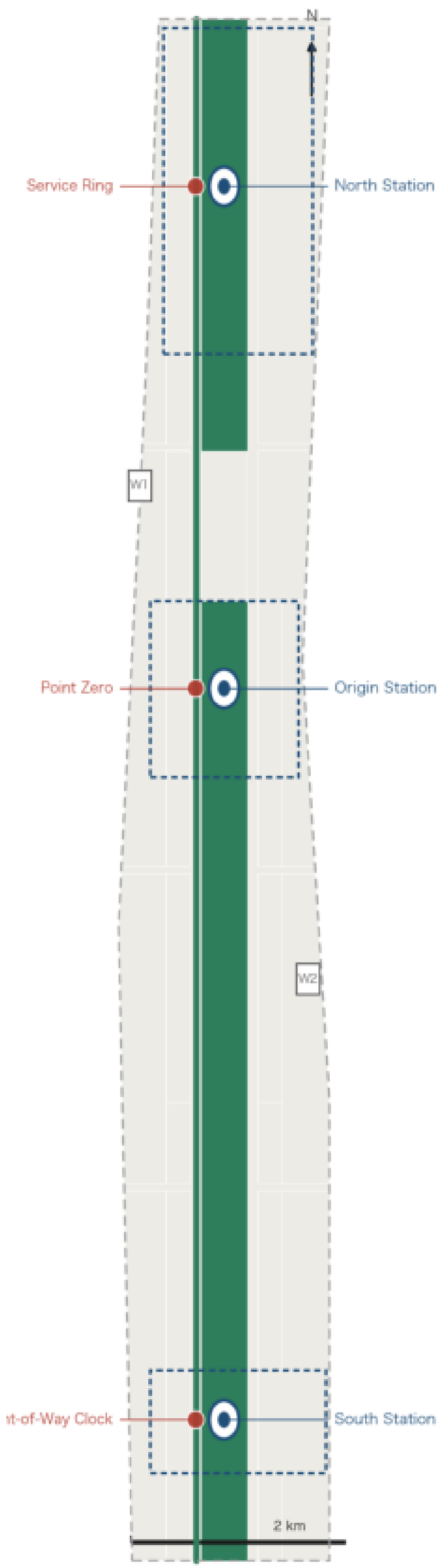
Submitted by qxj | Generated by Claude Opus 5 (claude-opus-5) in Claude Code

# Overall Spatial Structure: One Line, Three Stations, Two Wings

The trunk runs 9.72 km; one station per key area, mean spacing 3.89 km

## Overall Spatial Structure: One Line, Three Stations, Two Wings

GREEN:LIGHT — An Urban Corridor That Gives AI Scenarios a Green Light




### 01 Core proposition

The Jing-Zhang Railway was the first trunk line surveyed, designed and built by Chinese engineers, opened in 1909, carrying coal and people. It left behind a continuous linear green corridor of about 9.7 km running through all three key areas — the only continuous channel in this district that reaches from a trunk line all the way to residential doors.

The proposal turns it into an urban corridor where AI scenarios can take to the street legally, slowly, under supervision and with a way out, assembling the full set of facilities and rules that street entry requires. The green light is a metaphor for admission — and it can always be switched back to red.

### 02 Roles of the three stations

**North Station** — Zhongzhiyuan AI Acceleration Area

Full-stack AI innovation and governance voice | regional depot, vehicle servicing, compute access

**Origin Station** — Beijing AI Origin Community

World-class AI ecosystem | human-machine demonstration segment, public experience hub

**South Station** — Dazhongsi AI Industry Cluster

AI-native business formats | instant fulfilment for retail districts, night restocking

### 03 Two Wings

W1 **Zhongguancun Technology Service Wing**

Compute and capital interface: global factor allocation, Zhongguancun brand and capital support, feeding compute, funding and talent into the trunk

W2 **Xiaoyuehe Scenario Empowerment Wing**

Pilot segment: AI scenarios run closed here first and only open to the rest of the trunk after passing

### 04 Legend

- Green trunk line
- Urban blocks along the corridor
- AI Pilgrimage Landmarks
- To-Door Stations
- Key area extent
- Grey dashed line: provisional rough boundary, not a statutory re

### 05 Key metrics

| | |
|---|---|
| Trunk length | 9.72 km |
| Design area | 11.41 sq km |
| Key areas | 3 |
| Landmarks | 3 |
| To-door points | 32 |
| Green ratio | 24.7% |

Source: provisional rough boundary derived from the announcement's textual extent (provisional_rough); not usable as a statutory redline or a precise-area basis. Geometry and areas computed in EPSG:4548. All spatial suggestions are conceptual proposals for professional teams to develop further, not approved conclusions.

## Division among the three stations

North Station carries the regional depot, vehicle servicing and compute access; Origin Station is the human-machine demonstration segment and public experience hub; South Station serves instant fulfilment for retail and night restocking.

## Input from the two wings

The Zhongguancun Technology Service Wing is the compute and capital interface; the Xiaoyuehe Scenario Empowerment Wing is the pilot segment where scenarios run closed first and only open outward after passing the gates.

## The coordination loop

The three areas and two wings form a loop: the wings supply factors, the three areas divide the load, the trunk connects them and delivers results to the door, and operating data flows back to improve the rules.

## Boundary

The provisional boundary is about 0.11% from the announced 11.4 sq km: the area is usable, the shape is rough, and it is not a statutory redline or a precise-area basis.

# Land-Use Structure and Building Scale

A topology-safe partition: category areas total exactly the site area, with no gaps or overlaps

## Land-Use Structure: Partition Along the Continuous Green Trunk

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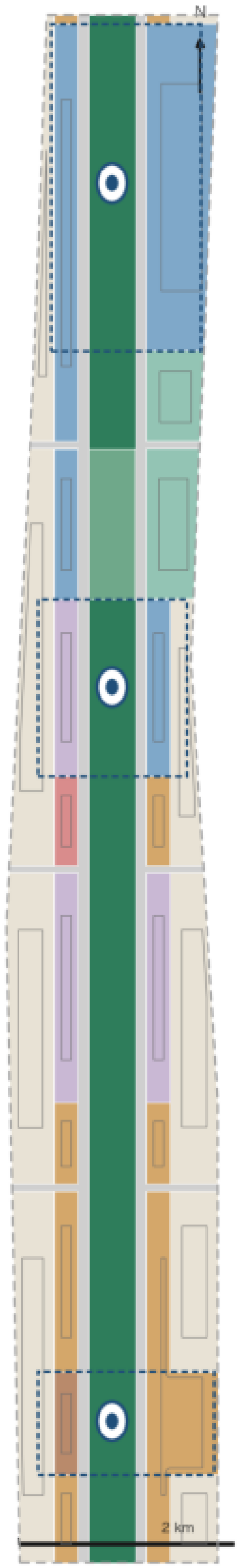


### 01 Land-use area and share

| Code | Category | Area (ha) | Share |
|---|---|---|---|
| 0701 | Urban housing | 323.4 | 28.3% |
| 1401 | Park green space | 254.8 | 22.3% |
| 0802 | Research | 155.4 | 13.6% |
| 1207 | Urban road | 143.0 | 12.5% |
| 05 | Commercial services | 107.2 | 9.4% |
| 0702 | Community services | 61.5 | 5.4% |
| 0804 | Education | 49.9 | 4.4% |
| 0805 | Sports | 27.4 | 2.4% |
| 0803 | Culture | 9.9 | 0.9% |
| 0806 | Medical | 8.7 | 0.8% |
| | Total | 1,141.3 | 100.0% |

### 02 Partition method and design intent

A topology-safe partition, not hand-drawn blocks

A continuous green trunk

a

The commonly missed item: charging and back-of-house

i

What is deliberately not given

o

Source: provisional rough boundary derived from the announcement's textual extent (provisional_rough); not usable as a statutory redline or a precise-area basis. Geometry and areas computed in EPSG:4548. All spatial suggestions are conceptual proposals for professional teams to develop further, not approved conclusions.

### The banding logic

Active frontage on the corridor, existing housing behind. From the spine outward: trunk corridor, flanking streets, inner frontage band, outer existing blocks.

### The commonly missed item

Charging and back-of-house space is missed in almost every comparable proposal. Light vehicles need physical space for battery swapping, parking, maintenance and loading; it is placed in community service and research land, and treated as something to be seen.

### What is deliberately not given

No FAR caps, building heights, setbacks or parcel-level verdicts. Official control indicators are recorded as missing, pending formal planning conditions. The FAR and density reported are the scheme's own estimates.

### Retention as the default

The corridor is built out and the overwhelming majority of buildings are retained. Renovation is limited to inner-frontage ground floors and entrances, not primary structure; anything touching third-party space requires the owner's agreement.

# Detailed Design of the Three Key Areas

Drawn at one shared scale for direct comparison, sharing one component library

## Detailed Design of the Three Key Areas

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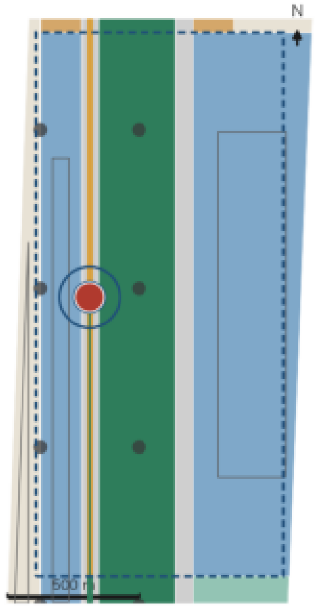
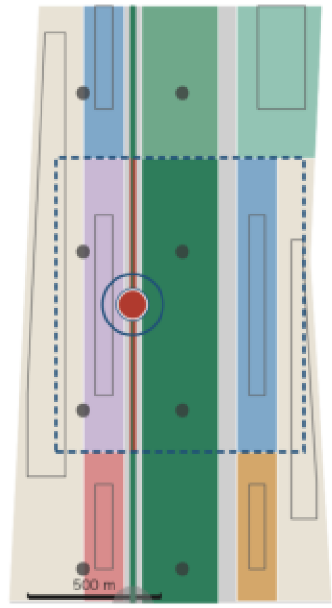
### Zhongzhiyuan AI Acceleration Area

192.1 ha (announced; provisional extent, not a precise-area basis)



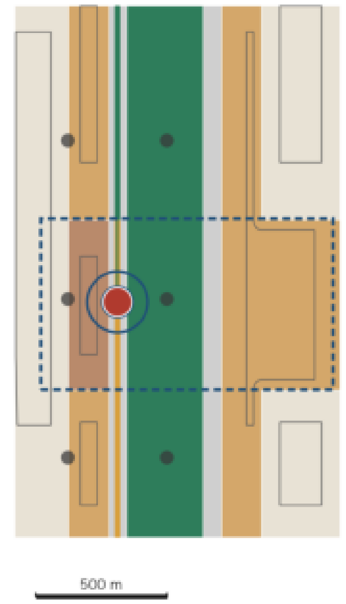

### Beijing AI Origin Community

104.3 ha (announced; provisional extent, not a precise-area basis)




### Dazhongsi AI Industry Cluster

72.0 ha (announced; provisional extent, not a precise-area basis)




### 01 Full-stack AI innovation | global governa

Role: regional depot, vehicle servicing and maintenance, compute access. The Service Ring landmark opens charging and maintenance to public view, making hidden facilities and frontline maintenance labour visible.

| | |
|---|---|
| Announced area | 192.1 ha |
| Trunk RoW grade | B |
| Opening phase | Phase 3 Open |

Provisional rough extent, not a statutory redline or precise-area basis. Spatial suggestions inside are conceptual, for

### 02 World-class AI innovation ecosystem

Role: human-machine demonstration segment, public experience hub. Pedestrian flow peaks here, so the trunk is a Grade C bottleneck where light vehicles are barred at peak and rerouted. Point Zero publicly displays the day's real results, including failure counts.

| | |
|---|---|
| Announced area | 104.3 ha |
| Trunk RoW grade | C |
| Opening phase | Phase 2 Guided |

Provisional rough extent, not a statutory

### 03 AI-native business formats

Role: instant fulfilment for retail and consumption, night restocking. The Right-of-Way Clock shows current-hour priority in three colours, echoing the historic bell culture of Dazhongsi: a bell has always governed time, and time-sliced right-of-way governs exactly that.

| | |
|---|---|
| Announced area | 72.0 ha |
| Trunk RoW grade | B |
| Opening phase | Phase 3 Open |

Provisional rough extent, not a statutory

### 04 Shared public-space component library across the three key areas

The same standardised modules are reused across the three key areas and all to-door points, keeping interfaces consistent, maintainable and extensible. All components are conceptual suggestio

**Pick-up module**
Shared pick-up lockers for parcels, meals, medicines and test samples, at accessible height

**Boarding module**
Low-speed shuttle stop and waiting area, step-free access, tactile paving and audio cues

**Charging module**
Battery swap and charging bays, night noise and lighting limits, service access separated from footways

**Right-of-way display**
Shows current-hour priority in the same visual language as the Right-of-Way Clock, legible at a glance

**Curb-cut module**
Continuous ramps designed for wheelchairs that also serve trolleys, luggage and cycles: an interface designed for the most vulnerable serves everyone

**Monitoring module**
Pedestrian density counting with no facial or identity recognition; auto-clears vehicles above thresholds

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## Zhongzhiyuan: bring the back-of-house forward

Regional depot, vehicle servicing and compute access. The Service Ring opens charging and maintenance to public view. The trunk here is Grade B, in Phase 3.

## Origin Community: bring results forward

The human-machine demonstration segment and public experience hub. Flow peaks here so the trunk is a Grade C bottleneck, barred at peak and rerouted. Point Zero publishes the day's real results including failures. Phase 2, with marshals.

## Dazhongsi: bring the rules forward

Instant fulfilment for retail and night restocking. The Right-of-Way Clock shows current-hour priority in three colours, echoing the historic bell. Grade B, Phase 3.

## Six modules in the component library

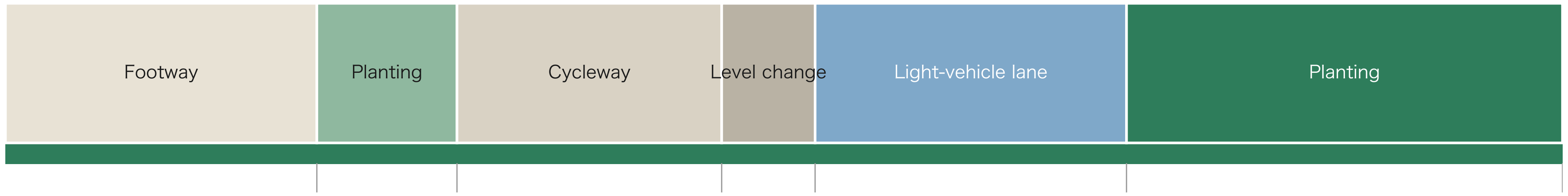
Pick-up, boarding, charging, right-of-way display, curb cut, monitoring. The curb-cut module is mandatory: a ramp designed for wheelchairs ends up serving trolleys, luggage and bicycles as well.

## 04 Graded Right-of-Way and Cross-Section Width Allocation

Physics first, then speed, then behaviour, and only then time. There are no measured cross-sections; the following shows relative proportions only, with absolute widths pending survey.

### A Grade A: ample width

5.30 km

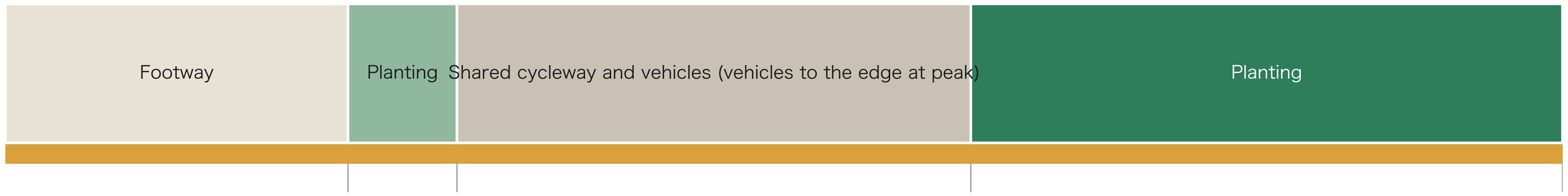



Available corridor width (relative proportion, absolute pending survey)

Full-time physical separation; each mode has its own lane, no time slicing

### B Grade B: moderate width

3.31 km

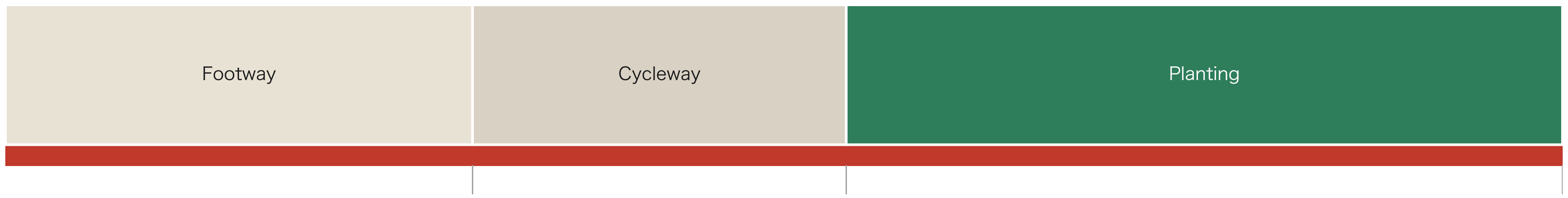



Available corridor width (relative proportion, absolute pending survey)

Normally separated; at peak, light vehicles are demoted to the edge and give up the main channel

### C Grade C: bottleneck, insufficient width

1.11 km




Available corridor width (relative proportion, absolute pending survey)

Light vehicles barred at peak and rerouted; entered on the minor-works list, with the treatment confirmed after a cross-section survey

Time slicing applies only to Grade B and C segments at peak: 07:00 - 09:30 people first, 09:30 - 17:00 vehicles and delivery first, 17:00 - 21:00 people first, 21:00 - 07:00 freight first. Grade A accounts for 54.5% of the trunk, which means more than half of it needs no time-slicing rule at all. The yielding covenant is the third line of defence: vehicles always yield and must visibly signal awareness — predictability over avoidance rate.

# Graded Right-of-Way and the Blue-Green System

Time slicing is the fourth line of defence, not the primary means

## Graded Right-of-Way and the Blue-Green System

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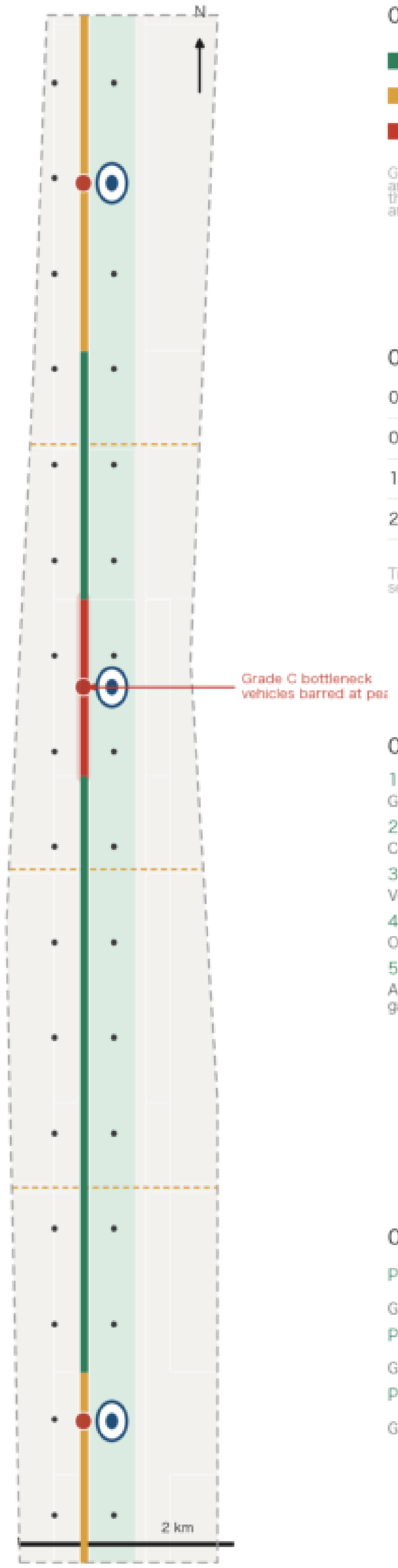


### 01 Right-of-Way Grades (inferred widths, pending survey)

| | |
|---|---|
| Grade A: full-time physical separation, no time slicing needed | 5.30 km |
| Grade B: normal separation; peak-hour vehicles demoted to the edge | 3.31 km |
| Grade C: bottleneck; vehicles prohibited at peak, rerouted | 1.11 km |

Grade C bottlenecks account for 11.4% of the trunk length. Grades are inferred from estimated widths and pedestrian density; no measured cross-sections exist in the repository or public sources. What the proposal asserts is the grading method itself; grades will be recomputed once survey data arrives, leaving the method intact.

### 02 Time-Sliced Right-of-Way (B and C segments only)

| | | |
|---|---|---|
| 07:00 - 09:30 | People first | Speed-capped, edge lane only, always yield |
| 09:30 - 17:00 | Vehicles and delivery first | Main channel available, low pedestrian flow |
| 17:00 - 21:00 | People first | Vehicles leave the main channel |
| 21:00 - 07:00 | Freight first | Main channel available, noise and lighting limits |

Time slicing is the fourth line of defence, not the primary means. Wherever width allows, lanes are separated physically; time slicing applies only to narrow Grade B and C segments at peak hours.

### 03 Five Layers of Human-Machine Coexistence

1 Physical separation
Grade A/B/C by cross-section width; separate physically wherever width allows

2 Speed and size
Collision harm scales with speed squared; a speed cap beats any avoidance algorithm

3 The yielding covenant
Vehicles always yield and must visibly signal awareness; predictability over avoidance rate

4 Time slicing
Only for B/C segments at peak; rules must be legible at a glance (Right-of-Way Clock)

5 Fail-safe and exit
Auto-clear above density thresholds, universal stop button, every event attributable, roll back when gates fail

### 04 Phasing is a risk-graded opening sequence, not a build schedule

| | | |
|---|---|---|
| Phase 1 Closed | Xiaoyuehe pilot segment | 257.3 ha |
| Gates: encounters per km-hour, yield success, emergency-stop rate \| roll back to closed | | |
| Phase 2 Guided | Origin Station segment, with marshals | 271.3 ha |
| Gates: complaints, remote takeovers, comfort survey \| roll back to Phase 1 | | |
| Phase 3 Open | North and south segments, full line | 612.8 ha |
| Gates: bottleneck share, point coverage \| any segment failing returns to Phase 2 | | |

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## Five layers of defence

Physical separation, speed and size, the yielding covenant, time slicing, then fail-safe and exit. The order matters: relying on rules for safety is the weakest approach available.

## The fail-safe

Density counting with no facial or identity recognition, clearing vehicles automatically above the threshold; a stop button anyone can press; every event attributable; rollback when gates fail.

## The corridor

The green system totals 282.2 ha, a green ratio of 24.7%. The trunk runs continuously, the physical precondition for the whole proposal. How stitching streets cross it is conceptual; no engineering conclusion is offered.

## One shortcoming stated openly

To-door points within 300 m cover only 56.0% of residential land. The 81.7% at 500 m looks better, but the corridor is only about 1.3 km wide so that measure approaches saturation. The conclusion: points must be densified outward per the 300 m gap.

# 06 Three AI Pilgrimage Landmarks: Three Kinds of Invisibility

AI is invisible: compute sits in machine rooms, models in the cloud, code on screens. The conventional answer is an exhibition hall — passive, enclosed, showing a promotional film. Each of these three landmarks addresses o

## 01 Point Zero

### Results are invisible

The public cannot check what AI actually achieved

A working pick-up point that publishes the day's real results across every scenario on the line, including failure counts. It shows them when the numbers look bad too; otherwise "verifiable" is an empty word.

Location: Origin Station

## 02 Right-of-Way Clock

### Rules are invisible

Residents do not know who this path belongs to right now

Shows current-hour priority in three colours for people, machines and freight. It belongs at Dazhongsi because a bell already stands there: a bell has always governed time, and time-sliced right-of-way governs exactly that. It is a governance interface that happens to take the form of a landmark.

Location: South Station

## 03 Service Ring

### Facilities and labour are invisible

Logistics back-of-house is the most hidden space in a city

Charging and maintenance back-of-house, opened to public view. Mechanical operation is genuinely interesting to watch, and frontline maintenance labour deserves to be seen: this is also the practical basis for the first of the seven user personas.

Location: North Station

Together they deliver three things

| | |
|---|---|
| Results visible → trust | The public can check, including failures |
| Rules visible → governance | Residents see at a glance who has priority now |
| Facilities and labour visible → respect | Frontline maintenance labour is acknowledged |

The reason to make the trip is not a photo opportunity but to watch a system that is genuinely running — and to see, on the spot, where it still falls short. All three are conceptual suggestions for professional teams to develop, constitute no approved conclusion, and are

# 07 AI Scenario Cards and User Personas

Twelve cards across six categories. Each fixes seven fields: location, participants, AI capability, data source and privacy boundary, human review point, measurable indicators, and failure and exit conditions. A scenario witho

## Distribution of the 12 cards


12 cards in total

## User personas (seven)

1 Frontline maintenance staff

2 Older people at home

3 AI start-up members

4 Commuters and parents

5 Wheelchair and visually impaired users

6 Last-stop shopkeepers

7 International researchers

Personas are defined by who actually uses this corridor every day, not by stacking demographic labels. Frontline maintenance staff come first, the practical basis for the Service Ring landmark.

## Three industrial test scenarios, all written as experiments

### Test 1 Time-sliced right-of-way pilot

Xiaoyuehe wing, Phase 1 closed, 257.3 ha

Control: before/after on the same segment plus a parallel Grade A control | Indicators: encounters per km-hour, yield success rate, mean yielding distance, emergency-stop rate | Review: every stop and complaint reviewed individually with a named responsible party | Exit: roll back to closed testing and suspend opening if gates fail

### Test 2 Queueing and dispatch at shared pick-up points

Origin Station

Control: points with dispatch logic against first-come-first-served | Indicators: mean waiting time, peak queue length, collection failure rate, comfort survey | Review: staff intervene in queue conflicts and requests for help | Exit: restore manual organisation if worse than the human baseline

### Test 3 Proof-of-delivery check and address validity

Real addresses line-wide

Control: automatic decisions in parallel with human sampling, comparing agreement | Indicators: valid-proof recognition rate, address corrections, error rate, dispute handling time | Review: all disputed cases decided by staff, automatic decisions never final | Exit: disable automatic decisions if the error rate exceeds the threshold. Boundary: house numbers and delivery surroundings only, no facial or identity recognition

# Phasing as a Risk-Graded Opening Sequence

Phasing is not a construction sequence. Each phase carries quantified gates and rollback conditions; Phase 1 sits in the Xiaoyuehe pilot segment where flow is not densest, proving things out before the hardest segment.

## Phase 1 Closed

Xiaoyuehe pilot segment

257.3 ha

Gate indicators: Encounters per km-hour, yield success rate, emergency-stop rate

Rollback: Roll back to closed testing and suspend opening

## Phase 2 Guided

Origin Station segment, with marshals

271.3 ha

Gate indicators: Complaint rate, remote takeover rate, comfort survey

Rollback: Return to the Phase 1 closed state

## Phase 3 Open

North and south segments, full line

612.8 ha

Gate indicators: Bottleneck share, to-door coverage

Rollback: Any failing segment returns to Phase 2 on its own

The three phase areas sum to the design area: the phasing layer is a complete partition of the boundary. This mechanism does not promise zero conflict; it promises that conflicts can be detected, attributed and exited from when gates are not met. Engineering feasibility is for professional teams to develop.

# Metrics and Evidence: What Is Computed and What Is Missing

47 of 52 metrics are recomputed from geometry; 5 official controls are reported as pending

## Metrics and Evidence: What Is Computed and What Is Still Missing

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### 01 Computed From Geometry

Each carries a formula, source files and a confidence level, independently recomputable from geometr

| | |
|---|---|
| Design area | 11.413 sq km |
| Partition total | 11.413 sq km |
| Trunk length | 9.72 km |
| Grade A length | 5.30 km |
| Grade B length | 3.31 km |
| Grade C length | 1.11 km |
| Bottleneck share | 11.4% |
| Green ratio | 24.7% |
| Public space ratio | 2.2% |
| Road land ratio | 12.5% |
| Building density (scheme estimate) | 19.5% |
| FAR (scheme estimate) | 1.26 |
| Total floor area (estimate) | 14.42 M sq m |
| To-door points | 32 |
| Coverage of housing, 300 m | 56.0% |
| Coverage of housing, 500 m | 81.7% |
| Mean station spacing | 3.89 km |
| Key areas total | 369.3 ha |
| Phase 1 closed pilot | 257.3 ha |
| Phase 2 guided | 271.3 ha |
| Phase 3 open | 612.8 ha |

### 02 Pending Official Data

Official control indicators are recorded as missing in the site package; the proposal makes nc

| | |
|---|---|
| ☐ FAR limit | Pending |
| ☐ Building height limit | Pending |
| ☐ Building density limit | Pending |
| ☐ Green ratio requirement | Pending |
| ☐ Setback requirement | Pending |

The FAR and building density reported are the conceptual scheme's own estimates, not planning control indicators, and imply no conclusions about building heights or retain/renovate/demolish decisions.

### 03 Disclosed assumptions

A-BOUNDARY-001
Provisional rough boundary, 0.11% off the announced area, not a statutory redline

A-STOREY-001
Storey counts are assumed, used for floor area and FAR estimates

A-WIDTH-001
No measured cross-sections; grading is a method demonstration

A-THRESHOLD-001
Speed, size, noise and density thresholds are advisory, pending professional review

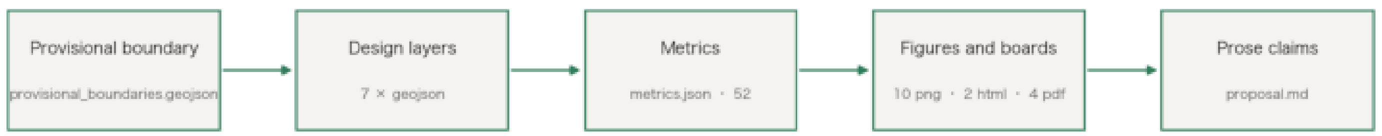
A-MODEL-001
Generation disclosure: model, scripts and scope of human review

A-HISTORY-001
Historic station correspondences serve narrative only, not spatial evidence

### 04 Local validation status

| | |
|---|---|
| Land-use topology | No gaps, no overlaps |
| Metrics schema | Valid |
| Area self-proof | Partition total = site area |
| Spatial review | Pass |

### 04 Evidence chain: from geometry to prose, recomputable at every step




Any number can be traced back and recomputed one step upstream. The prose does not repeat machine indexes; complete source, standard and depth coverage live in the structured files.

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## Area recomputation as self-proof

Category areas total exactly the site area. Not a coincidence but a necessary consequence of the method: disjoint cells clipped to one boundary must union back to it. That equality is itself the proof, and more reliable than a declaration.

## Recomputable

Each metric carries a formula, source files, a confidence level and its related assumptions; areas are computed in EPSG:4548, and green space, public space and footprints use unions rather than sums.

## Six disclosed assumptions

Provisional boundary precision, missing official controls, assumed storey counts, no measured cross-sections, advisory safety thresholds, generation disclosure, and unverified historic station correspondence.

## No fabricated certainty

Every statement touching companies, funding or policy is a citation of public information or a proposed mechanism, not a confirmed arrangement; the test scenarios are proposed validation work, not approved operations. The final judgement rests with people and professional teams.